UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #01

Course:	**CHE 433: PROCESS SIMULATION WITH ASPEN PLUS**
Term:	**Fall 2025**
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Student names: ______________________________________

Due Date: Friday, September 5th TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a typed report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Also place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Review of Chemical Engineering Fundamentals

Your first assignment after six months of training in the **UIC Specialty Chemicals Corporation** is to provide a material balance for a cyclohexane production unit and preliminary cost estimates of the utilities. A simple process flow diagram (shown in Figure A below) has been given to you by a mid-level engineer who has decided to leave the company. The data for this process are given below:

Benzene (BZ) Feed
Flow: 92.14 lbmole/hr
Temperature: 100 ^oF
Pressure: 15 psia

Hydrogen (H2) Feed
Flow: 283.8 lbmole/hr
Temperature: 120 ^oF
Pressure: 335 psia
H_2 mole fraction : 0.975
N_2 mole fraction : 0.005
CH_4 mole fraction : 0.020

Charge Heater
Temperature: 300 ^oF
Pressure: 330 psia

Reactor
Temperature: 400 ^oF
Pressure: 315 psia
Benzene Conversion: 99.86%
H_2 to BZ molar ratio: 3.3

High Pressure (HP) Product Separator (Flash Drum)
Temperature: 120 ^oF
Pressure : 300 psia

Low Pressure (LP) Product Separator (Flash Drum):
Pressure : 15 psia

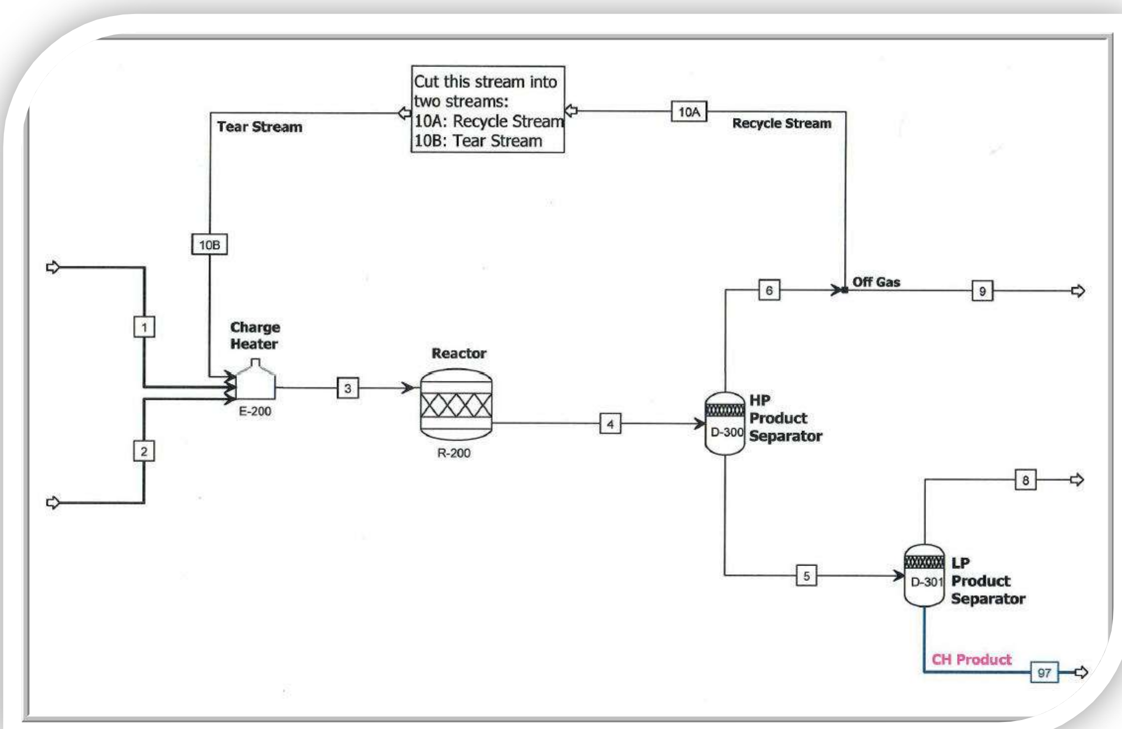
Your tasks

1. Complete the process flowsheet by adding the missing unit operations and provide a brief explanation for every unit you add.
2. Prepare the material balance in the format of the **ENG Calculations** of the **CHE Calculator**. Provide a brief explanation on how you calculate the effluent streams from each unit operation. Additionally provide a brief description of the numerical method(s) you have used to derive the material balance
3. Provide a preliminary cost of all utilities
4. Propose a method to reduce the utilities cost and do a simulation to calculate the cost reduction of your proposal

Data and Assumptions:

1. You may assume that the hydrocarbon mixtures with hydrogen presence in the process follow the Peng-Robinson equation of state
2. You may also assume that the K-values for the flash calculations can be calculated from Raoult's law or the Wilson equation
3. The fuel in the charge heater is natural gas
4. Utility cost data can be obtained from a variety of sources

Figure A: Hydrogenation of Benzene to Cyclohexane



ASPEN Report 1

Kyle Hankosky and Khoa Nguyen

September 8, 2025

1 Missing Unit Operations

We first place a pump on the Benzene stream S1 and a compressor on the tear stream S10B to increase their pressures to 335 psia, which helps to avoid a large pressure difference between streams S1, S2, and S10B. A mixer is then added to uniformly combine all three streams before going into the charge heater. As the species exits the reactor, we will add a heat exchanger before going to the HP separator to decrease the temperature to the separator's operating conditions. Lastly, a valve is placed in front of the LP separator, which helps to decrease the pressure to the LP separator's operating condition. In total, we have 5 additional unit operations, and an updated flow diagram is shown below:

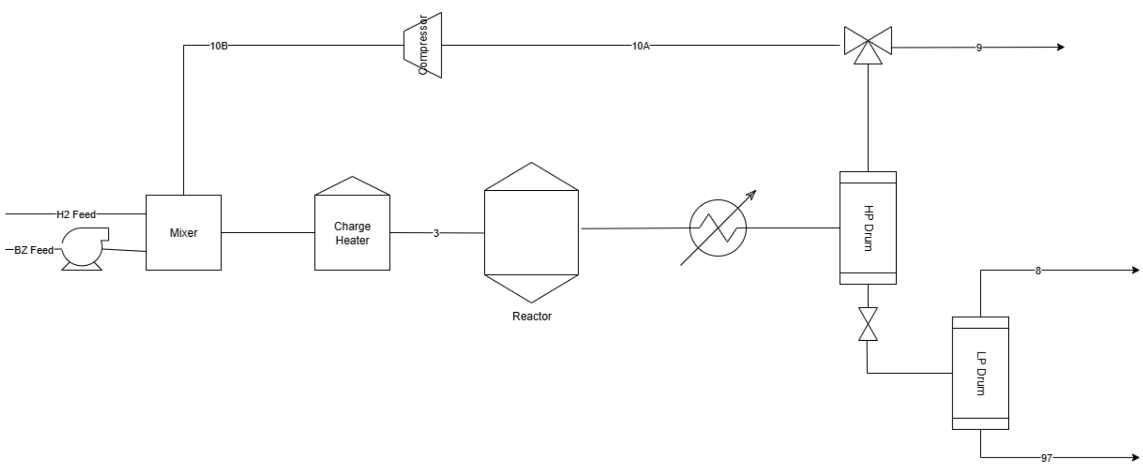


Figure 1: Updated Flow Diagram

2 Material Balance Using CHE Calculator

In order for us to complete the material balance of this process, we will use both the VLE and ENG calculations. The calculations are arranged by streams and unit operations in the ENG section, which correlates with the updated diagram and is shown below:

ENG Calculations	Peng-Robinson													
Molar Composition	S1	S2	Comp Out	Mix Out	Heater Out	Reactor Out	HE	HP5 Bottom	HP6 Top	10A (Recycle)	10B (Tear)	Valve	LP97 Top	LP97 Bot
C6H6														
H2														
N2														
CH4														
C6H12														
Sum														

Figure 2: ENG Calculations Layout

This particular process for cyclohexane manufacturing involves 5 species: Benzene (C_6H_6), Hydrogen (H_2), Nitrogen (N_2), Methane (CH_4), and Cyclohexane (C_6H_{12}). The overall chemical reaction is:

$$C_6H_6 + 3H_2 \rightarrow C_6H_{12}$$

The molar compositions of species for everything up to the heat exchanger can be calculated using simple algebra. Streams S1 and S2 are the Benzene and Hydrogen feeds, respectively, and their molar compositions are given. Both of their molar flows were given in lbmol/hr, which were kept in lbmol/hr for calculation purposes. Molar compositions for S10B (Comp Out) can't be determined until values for the HP separator and splitter have been calculated, so we assume 0 for all species initially.

To determine the molar compositions of species exiting mixer (Mix Out), we first calculate the molar flow at the exit of mixer, which is simply the sum of S1, S2, Comp Out. The molar composition of each species is then multiplied by S1, S2, Comp Out, and then divided by molar flow rate of Mix Out. Values for the exit of the charge heater (Heater Out) is identical to Mix Out since there are no reactions and additional streams.

To determine the compositions of species exiting the reactor (Reactor Out), we first calculate the molar flow of each species, which is identical to the heater, since there are no reactions and no additional streams. With the additional information that the conversion rate of Benzene is 99.86%, we first calculate the molar flow rate of each species, add everything up, which would be the total molar flow rate exiting the reactor, and each species molar flow rate is then divided by the total molar flow rate to get our molar compositions. We will call ξ as the extent of reaction with:

$$\xi = 0.9986 * S1$$

The molar compositions of species exiting reactor is calculated as follows:

Benzene (C_6H_6):

$$\frac{S1 - \xi}{Total molar flowrate}$$

Hydrogen (H_2):

$$\frac{0.975 * S2 - 3\xi}{Total molar flowrate}$$

Nitrogen (N_2):

$$\frac{0.005 * S2}{Total molar flowrate}$$

Methane (CH_4):

$$\frac{0.028 * S2}{Total molar flowrate}$$

Cyclohexane (C_6H_{12}):

$$\frac{\xi}{Total molar flowrate}$$

The molar compositions and flow rate of the heat exchanger (HE) is identical as Reactor Out since there are no reactions and additional streams.

To determine molar compositions and flow rates of the bottom stream of HP Separator (HP 5 Bottom) and top stream of the same unit (HP 6 Top), we will use VLE Calculations along with ENG Calculations since both vapor and liquid are present within this unit.

We first fill in the molar compositions of the HP Separator, which is identical to that of the Heater Out. Along with the k-values of each species, which were provided, we can easily calculate the Vapor to Feed Molar Ratio (α) using the Goal Seek function. After this, we set up a table consisting of Zi, Ki, Xi, Yi, with each symbol representing the molar composition, k-values, liquid fraction, and vapor fraction of each species, respectively.

Both Xi and Yi are calculated using the following formulas:

$$Xi = \frac{Zi}{1 + \alpha(Ki - 1)}$$

and

$$Yi = Xi * Ki$$

After calculating these values, molar compositions of HP 5 Bottom will be equal to Xi values, and HP 6 Top will be equal to Yi, assuming that all liquid flows to bottom stream and all vapor flows to top stream. The flow rates for HP 5 Bottom is then calculated using the formula:

$$1 - \alpha * FlowrateHE$$

And flow rate of HP 5 Top is calculated using the formula:

$$\alpha * FlowrateHE$$

The molar compositions and flow rate of the valve are identical to those of HP 5 Bottom since there are no reactions and no additional streams.

Similar to the HP separator, LP separator is also divided into a top section (LP 97 Top) and a bottom section (LP 97 Bottom). Since both vapor and liquid are present, we will once again use the VLE Calculations. The molar compositions of the LP Separator will be identical to those of the valve since there are no reactions and no additional streams. Using new k-values of each species for the LP separator, which were provided, we can once again calculate α using the Goal Seek function. Upon obtaining the value for α, we can calculate Xi and Yi for the LP Separator using the same formulas as those for HP separator. Using the same assumption that all liquid flows to the bottom section and all vapor flows to the top, the molar compositions for both LP 97 Bottom and LP 97

Top are obtained. Additionally, the flow rates for both top and bottom sections of the LP Separator is calculated using the same formulas as the ones for HP separator, with the exception that the α values used are exclusive only to that unit, i.g α value for LP separator can only be used for LP separator. α would also be multiplied by the flow rate of the valve instead of HE for the LP Separator.

Molar compositions of the recycle stream 10A are identical to those of HP 6 Top since there are no reactions and no additional streams. We will also be introducing a purge factor for the splitting between the recycle and tear stream, crucial in determining the flow rates of both, and is arbitrarily chosen as 0.02. With this purge factor, we can then calculate the flow rate of the recycle stream, which is simply:

$$(1 - PurgeFactor) * FlowRate\ HP6Top$$

Material balance on the tear stream is accomplished through 2 iterations of Newton method, the ratio between Hydrogen and Bezene, and the purge factor. Below is a complete material balance on all streams of this process.

ENG Calculations	Peng-Robinson													
Molar Composition	S1	S2	Comp Out	Mix Out	Heater Out	Reactor Out	HE	HP5 Bottom	HP6 Top	10A (Recycle)	10B (Tear)	Valve	LP97 Top	LP97 Bot
C6H6	1.0000	0.0000	0.00002794	0.1634	0.1634	0.000437642	0.000437642	0.001287	0.00002794	0.000027935	0.000027935	0.001287	0.0004	0.0013424
H2	0.0000	0.9750	0.14556813	0.5392	0.5392	0.098615430	0.098615430	0.001239	0.14556813	0.145568127	0.145568127	0.001239	0.0204	0.0000083
N2	0.0000	0.0050	0.23958006	0.0824	0.0824	0.163496395	0.163496395	0.005704	0.23958006	0.239580061	0.239580061	0.0057	0.0927	0.0001101
CH4	0.0000	0.0200	0.59516452	0.2085	0.2085	0.413724798	0.413724798	0.037432	0.59516452	0.595164518	0.595164518	0.0374	0.5898	0.0019275
C6H12	0.0000	0.0000	0.01965936	0.0066	0.0066	0.323725734	0.323725734	0.954338	0.01965936	0.019659358	0.019659358	0.9543	0.2999	0.9964012
Sum	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0033	0.9998
Stream Properties	S1	S2	Comp Out	Mix Out	Heater Out	Reactor Out	HE	HP5 Bottom	HP6 Top	10A (Recycle)	10B (Tear)	Valve	LP97 Top	LP97 Bot
Molar Flow, lbmol/hr	92.1400	283.8000	188.0517	563.9917	563.9917	284.2414	284.2414	92.4684	191.7729	188.0517	188.0517	92.4684	5.584529233	86.8839

Figure 3: Completed Material Balance

3 Preliminary Cost of Utilities

The following costs of utilities were provided:

Cooling water: \$0.354/GJ

Natural gas: \$6.0/GJ

Electricity: \$0.06/kWh

All cost calculations for utilities will be performed on the basis of 8000 working hours.

To determine the utility cost of this process, we will first need to calculate the power and energy of every unit. For convenience purposes, we will calculate only the main type of power that each unit uses, which are:

Pump-electricity, Compressor-electricity, Heater-gas, Heat Exchanger-water, and Valve-electricity

All that is necessary for this section is the difference in enthalpies between the inlet and outlet of each unit, which can be obtained by utilizing the energy balance as well as the functions "Enthalpy" and "ZComp" from the CHE Calculator. The energy balance and cost analysis of the 5 aforementioned units are shown below:

Molar Enthalpy, kJ/kmol	Pump In (S1)	Pump out (Elec)	Comp in (10A)	Comp Out (Elec)	Heater In	Heater Out (Gas)	HE in	HE out(water)	Valve In	Valve Out (elec)
Temperature, C	37.7778	38.9641	48.8889	50.3295	45.7940	148.8889	204.4444	48.8889	48.3333	44.5274
Pressure, bar	1.0342	23.0974	22.7526	23.0974	23.0974	22.7526	21.7184	21.7184	22.7526	1.0342
Enthalpy Flow, kJ/s	51832.04	52110.37	-46436.02	-46386.81	-175.23	159.03	-2102.92	-2538.76	-147590.87	-148296.12
Duty [Power], kJ/s [kW]		3.23		1.17		334.27		2103.87		8.2514

Figure 4: Completed Energy Balance

	Energy consumption	Annual Utilities Cost, $
Pump Power, kWh	0.0009	$1,550.33
Compressor Power, kWh	0.00032	$559.79
Heater Energy, GJ	0.00033	$57,761.63
Cooling Water, GJ	0.0004	$4,443.49
	Total Cost	$64,315.25

Table 1: Energy Consumption and Annual Utilities Cost

4 Proposal for Utilities Cost Reduction

From the provided list of utility costs, we can see that the price of natural gas for the charge heater is the most expensive compared to that of cooling water and electricity. Because of this, we could remove the charge heater and install a heat exchanger at the compressor outlet. This would not only reduce our utility costs since water is free of cost, but it would also heat the inlet of the reactor to an appropriate temperature before the reaction starts.